
**Diesel Engines
V 1163 TB 73
Maintenance Schedule
M050396/22E**

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation. For this purpose, ensure, for instance:

- ensure that sufficient fuel is available
- ensure that the combustion air is dry and clean
- use only dry, clean compressed air
- raw water must be clean and filtered

Moreover, it is essential that

- trained personnel be used
- suitable tools be used
- genuine MTU spare parts and fluids and lubricants as approved in the current MTU Fluids and Lubricants Specification A001061 be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic parts from oil and fuel, never use organic agents to clean. Wipe with dry cloth only.
- Always replace seals/gaskets, do not re-use.

MTU maintenance system

The MTU maintenance concept is based on preventive maintenance and consists of Maintenance Levels W1 to W6 as outlined below.

Preventive maintenance permits advance operational planning and increases availability.

Main features of maintenance work on MTU engines

Maintenance level W1: Operational checks

W2, W3 and W4: Periodic maintenance services to be performed during out-of-service periods without the need for engine disassembly

W5: This maintenance level requires partial engine disassembly.

W6: Major overhaul This level requires complete engine disassembly.

The time intervals according to which W2 to W6 maintenance operations and the relevant inspection and maintenance tasks are to be performed are based on operational experience. They have been determined to ensure correct engine operation until the next maintenance work.

If a maintenance predictor is used, the operating hours indicated in the maintenance frequency chart can change.

Particular operating conditions may require the Maintenance Schedule to be altered to compensate.

NOTE:

The codes on the following pages indicate the groups and subgroups in the Engine Operating Instructions (Group G - Task Description) in which the tasks involved are described.

This maintenance schedule may include components which are not installed on your engine; these may be disregarded.

Application

1B: Fast vessels with high load factors

Load Profile

Load referenced to fuel-stop power	%:	100	85	15
Percentage of total operating time	%:	3	82	15

Maintenance Frequency Chart

Maintenance level	W1:	Operational checks, daily	X
	W2:	Operating hours	500
		Limit (months)	6
	W3:	Operating hours	1 000
		Limit (years)	1
	W4:	Operating hours	4 000
		Limit (years)	2
	W5:	Operating hours	8 000
		Limit (years)	6
Intermediate check		Operating hours	12 000
		Limit (years)	8
	W6:	Operating hours	24 000
		Limit (years)	18

Maintenance task schedule -overall system-

See Page 4

Maintenance Task Schedule -Engine-

Code No.	One-time additional services after the first 50 operating hours – on a new engine, or after W5 or W6 maintenance	
G00.049	Attachments	Check tightness of securing screws
G06.101	Valve gear	Check and adjust valve clearance
G10.111	Intake air line	Remove protective grill (only with intake from atmosphere)
G12.311	Fuel prefilter	Flush, replace filter elements
G12.321	Fuel duplex filter	Replace filter elements
G14.019	Engine coolant system	Clean strainers before coolant thermostat, remove mesh
G16.002	Engine oil	Take sample and analyze
G19.011	Engine mounts	Check buffer clearances

Code No.	Maintenance Level W1: – Operational checks	
G10.051	Exhaust system	Check exhaust-gas color Drain condensate (if drain provided)
G10.181	Intake air line	Check condensate drain lines for water discharge and obstructions
G12.311	Fuel prefilter	Check pressure differential
G14.011	Engine coolant	Check level
G16.002	Engine oil	Check level
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check running noises Check engine and external piping for leaks Check speed, pressures, and temperatures (where gauges are installed)
G86.051	Compressed air system	Drain condensate